



Biological, Technical, and Financial Feasibilities Study of Zero Water Discharge (ZWD) System Application in Low Salinity White Shrimp (*Litopenaeus vannamei* Boone) Urban Aquaculture. Study Case: Gresik District, North Coastal Area, East Java Province, Indonesia

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ABSTRACT

The research was aimed to analyze low salinity white shrimp production feasibility in urban areas using zero water discharge (ZWD) system. The research was set conducted by following five constitutive steps: (1) ZWD system installation and white shrimp post-larvae acclimatization to low salinity water (5 ppt), (2) White shrimp cultivation using low salinity ZWD system at three different stocking densities (200 ind/m³, 300 ind/m³, and 400 ind/m³) in 20 m³ ponds, (3) Biological feasibility and water quality analysis, (4) Technical feasibility analysis, and (5) production scheme development and financial feasibility analysis of three different stocking densities to produce 1000 kg shrimp/cycle. The best biological performance was achieved at 400 ind/m³ stocking density with survival rate, feed conversion ratio, specific growth rate and total biomass values were: 70.59±6.15%, 1.14±0.14, and 4.40±0.25 % BW/day, and 44.13±4.44 kg, respectively. The best pH level (7.61-8.27), DO (4.9-8.5 mg/L), temperature (29.3-30.1°C), NH₄⁺ level (0.0-0.5 mg/L), NO₂⁻ level (0.0-5.0 mg/L), and NO₃⁻ level (5.0-30.0 mg/L) were monitored at 200 ind/m³ stocking density. By considering several parameters: easiness of water sources, proper biophysical condition, and land topography structure, easiness of market access and raw material supplier, technically ZWD system for shrimp production was feasible at north coastal areas of East Java Province. Production scheme development needed six operational units consisted of (1) purchasing and inventory, (2) water and wastewater treatment, (3) shrimp production, (4) harvesting, (5) transaction and archive, and (6) marketing and distribution. To produce 1000 kg shrimp/cycle in 70 days culture period, 2-3 human resources and 1000-1250 m² production area were required. Based on financial analyses, the use of low salinity ZWD system was feasible at 300 ind/m³ and 400 ind/m³ stocking density. It was based on positive NPV Rp 47,593,537 and Rp 69,439,955, and higher IRR value 13.40% and 15.49% (more than discounted factor). It could be concluded that the application of low salinity ZWD system for white shrimp production in urban area, north-coastal area of Gresik, East Java province was biologically, technically, and financially feasible at 400 ind/m³ and 300 ind/m³ stocking densities.

Key words: *Litopenaeus vannamei* Boone, low salinity, Zero Water Discharge, technical feasibility, financial feasibility, urban aquaculture.

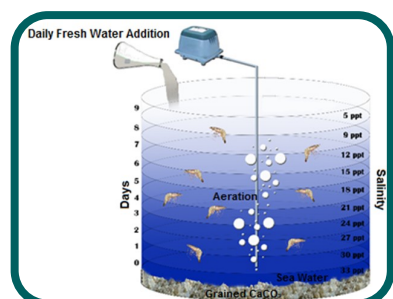
INTRODUCTION

Indonesia shrimp production in 2014 was 623.000 metric tons, which was dominated by white shrimp species (*Litopenaeus vannamei*) approximately up to 69.52% of total production (DJPB, 2015). The production actually does not ensure the sustainability of the particular industry, because most shrimp farms use conventional culture technology (typically a batch or a flow-through system). Besides, the system is considered as not environmentally friendly, because toxic effluent water can pollute aquatic environment nearby cultivation site. In term of space requirement, the system needs large production area and should be closed to coastal area (sea water access). It will contribute to the inflexibility of shrimp culture and shrimp industry sustainability in the near future.

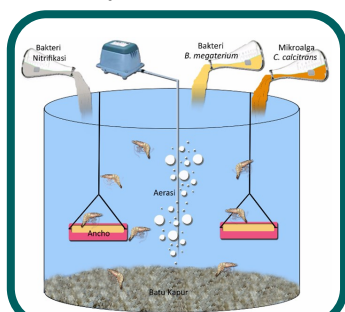
One cultivation strategy, Zero Water Discharge (ZWD), was developed to resolve these problems (Suantika *et.al*, 2015). ZWD principle is to reduce water usage, by implementing microbial consortium which recycles nitrogen compound in culture water, to clean harmful nitrogen substance and therefore water is reused (Suantika *et.al*, 2015). Based on Suantika *et al.* (2015), by cultivating white shrimp using ZWD system on laboratory scale, excellent shrimp culture performance reflected by acceptable water quality, 90% survival rate, and low FCR were obtained. For further development of the system on industry scale, a research on biological, technical, and financial feasibility need to be conducted before implementation of this low salinity ZWD system in a real aquaculture operation.

MATERIALS AND METHODS

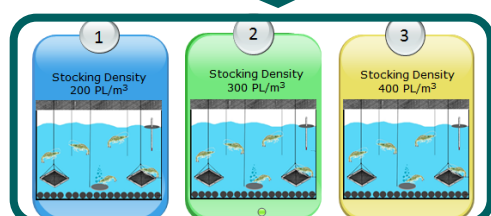
White shrimp acclimatization



ZWD System Installation



White Shrimp Cultivation using Low Salinity In Three Different Stocking Densities



Biological Feasibility Analysis

Technical Feasibility Analysis

Production Scheme Development

Financial Feasibility Analysis

Figure 1 Research Methods

RESULT AND DISCUSSION

Biological Feasibility Analysis

Based on biological parameters (Table 1), SD3 has the best production performance compared to SD1 and SD2. SD3 produced the largest total biomass and harvested shrimps are homogenously large (100 ind/kg).

Table 1. Biological Feasibility Analysis

Parameters	Treatments		
	SD1 (200 ind./m ³)	SD2 (300 ind./m ³)	SD3 (400 ind./m ³)
SR (%)	93.52±3.32 ^a	79.11±5.81 ^b	70.59±6.15 ^b
Total Biomass (kg)	27.75±1.55 ^a	36.25±3.01 ^b	44.13±4.44 ^c
FCR	1.05±0.07 ^a	1.06±0.08 ^a	1.14±0.14 ^a
SGR (%)	4.64±0.14 ^a	4.22±0.24 ^a	4.40±0.25 ^a
Size Distribution (%)	100-150/kg	86.77 ^a	91.93 ^{ab}
	150-250/kg	13.23 ^a	8.07 ^{ab}
			4.15 ^b

Technical Feasibility Analysis

According to those criterias, north coastal areas in East Java such as Tuban, Lamongan, Gresik, Sidoarjo, Pasuruan, Probolinggo, and Situbondo were suitable for shrimp urban aquaculture with ZWD system. The reasons were consisted of; (1) Areas are located close to sea water sources, (2) Provide suitable physical condition for shrimp growth, (3) Land topography is relatively flat, making it easier to build a farm, (4) Areas are located close to domestic market in main cities of East Java province; Surabaya and Lamongan, and (5) North coastal areas and its surrounding are free from any conflict.

Production Scheme Development

To operate white shrimp farming using ZWD system, there were six units identified: (1) purchasing and inventory, (2) water and wastewater treatment, (3) shrimp production, (4) harvesting, (5) transaction and archive, and (6) marketing and distribution (Figure 2).

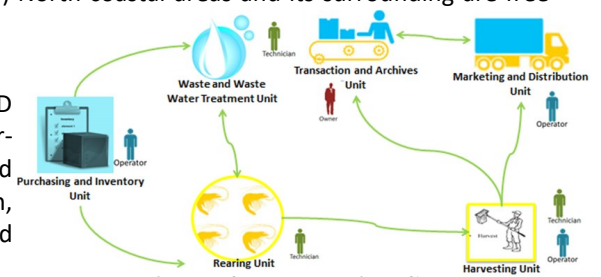


Figure 2. Production Scheme

Financial Feasibility Analysis

It can be clearly seen that SD2 and SD3 were financially feasible because have positive NPV, IRR value that higher than discount factor (around 13.40% and 15.49%), and B/C ratio that higher than 1 (around 1.13 and 1.22).

Table 2. Financial Ratio Calculations

Parameter	Treatments		
	SD1 (200 ind./m ³)	SD2 (300 ind./m ³)	SD3 (400 ind./m ³)
Investment Cost	Rp 443,520,000	Rp 358,080,000	Rp 318,230,000
Revenue	Rp 60,000,000	Rp 60,000,000	Rp 60,000,000
Production Cost/Cycle	Rp 49,087,750	Rp 43,494,625	Rp 44,227,125
Profit/Cycle	Rp 10,912,250	Rp 16,505,375	Rp 15,722,875
Profit/kg Shrimp	Rp 10,912	Rp 16,505	Rp 15,773
BEP (kg)	4,498	2,635	2,804
NPV	-175,315,390	47,593,537	69,439,955
B/C Ratio	0.605	1.13	1.22
Payback Period (year)	10.16	4.37	4.02
IRR (%)	<10	13.40	15.49

CONCLUSION

Based on this research, the implementation of ZWD system in shrimp urban aquaculture was feasible to be applied, especially with 400 ind./m³ and 300 ind./m³ stocking density in north-coastal area of West Java Province, such as Gresik.

ACKNOWLEDGMENT

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